

Social Media Marketing Campaign Analysis - Python Code

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# =====
# SOCIAL MEDIA MARKETING CAMPAIGN ANALYSIS
# =====
# Import Libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
# =====
# CREATE DATASET
# =====
data = {
    "Platform": ["Instagram", "Facebook", "YouTube", "Twitter", "LinkedIn"],
    "Ad_Spend": [20000, 15000, 30000, 10000, 12000],
    "Revenue": [60000, 40000, 90000, 22000, 25000],
    "Clicks": [4200, 3000, 6500, 1800, 2000],
    "Conversions": [450, 300, 700, 120, 150]
}
# Create DataFrame
df = pd.DataFrame(data)
# =====
# DISPLAY DATA
# =====
print("\n==== DATASET =====")
print(df)
print("\n==== DATASET INFO =====")
print(df.info())
# =====
# KPI CALCULATIONS
# =====
# ROI
df["ROI"] = ((df["Revenue"] - df["Ad_Spend"]) / df["Ad_Spend"]) * 100
# CTR
df["CTR"] = (df["Clicks"] / 10000) * 100
# Conversion Rate
df["Conversion_Rate"] = (df["Conversions"] / df["Clicks"]) * 100
print("\n==== DATA WITH KPI =====")
print(df)
# =====
# BEST PLATFORM
# =====
best_platform = df.loc[df["ROI"].idxmax(), "Platform"]
print("\nBest Performing Platform:", best_platform)
# =====
# VISUALIZATION
# =====
sns.set(style="whitegrid")
# Revenue Chart
plt.figure(figsize=(8,5))
sns.barplot(
    x="Platform",
    y="Revenue",
    data=df
)
plt.title("Revenue by Platform")
plt.xlabel("Platform")
plt.ylabel("Revenue")
plt.show()
# ROI Chart
plt.figure(figsize=(8,5))
sns.barplot(
    x="Platform",
    y="ROI",
    data=df
)
plt.title("ROI Comparison")
plt.xlabel("Platform")
plt.ylabel("ROI (%)")
plt.show()
# Correlation Heatmap
plt.figure(figsize=(8,5))
sns.heatmap(
    df.corr(numeric_only=True),
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        annot=True,
        cmap="coolwarm"
    )
    plt.title("Correlation Heatmap")
    plt.show()
    # =====
    # INSIGHTS
    # =====
    print("\n===== INSIGHTS =====")
    print("""
1. YouTube generated highest revenue.
2. Instagram achieved strong ROI.
3. Higher ad spend increases revenue.
4. Better conversion rate improves campaign success.
""")
    # =====
    # EXPORT CSV
    # =====
    df.to_csv("marketing_campaign_analysis.csv", index=False)
    print("\nCSV File Exported Successfully")
    # =====
    # END
    # =====

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